

DATA SCIENCE

RetailPulse — Customer Behavior Analytics & Revenue Intelligence Platform

Build a full-cycle data science project analyzing retail transaction data to segment customers, predict churn, forecast revenue, and deliver findings as an interactive executive business intelligence dashboard.

■ Domain Overview

Domain: Data Science is the discipline of extracting strategic business insights from large, complex datasets using statistical analysis, machine learning, and data visualization. Data scientists at companies like Amazon, Netflix, and Walmart build customer segmentation models, demand forecasting systems, and churn prediction pipelines that directly influence billions of dollars in business decisions. Enterprise data science demands rigorous hypothesis testing, reproducible analysis, and the ability to translate quantitative findings into clear business recommendations.

Objective: Conduct a full-cycle data science project on a retail transaction dataset: perform exploratory data analysis and data cleaning, apply RFM (Recency, Frequency, Monetary) customer segmentation with K-Means clustering, build a churn prediction classifier (>80% AUC), develop a revenue forecasting model using time series methods, perform cohort retention analysis, and deliver all findings in an interactive Streamlit business intelligence dashboard with executive-level commentary.

Expected Outcome: A fully reproducible data science project — Jupyter notebooks, cleaned dataset, trained model files, and an interactive Streamlit dashboard — covering EDA, customer segmentation, churn prediction, revenue forecasting, and cohort analysis, with a final business intelligence report presenting actionable recommendations.

■ Tech Stack

Category	Technology
Language	Python 3.12
Data Wrangling	Pandas 2.1, NumPy
Visualization	Matplotlib, Seaborn, Plotly Express
Machine Learning	scikit-learn 1.4 (clustering, classification, preprocessing)
Time Series	statsmodels (SARIMA) or Prophet (Facebook)
Dashboard	Streamlit 1.35
Statistical Testing	scipy.stats (t-test, chi-square, ANOVA)

Experiment Tracking	MLflow (model logging and artifact storage)
Notebook Environment	JupyterLab
Dependency Management	pip + requirements.txt or Poetry

■ Architecture & Data Flow

- Data Ingestion: Load raw CSV transaction data; validate schema; document source data dictionary
- EDA Notebook: Distribution analysis, outlier detection, missing value audit, correlation heatmaps, business KPIs computation
- Feature Engineering: RFM score computation, date-derived features (day-of-week, seasonality), customer lifetime features
- Segmentation Pipeline: K-Means clustering on RFM features; elbow method and silhouette score for K selection; cluster business profiling
- Churn Prediction: Binary label engineering, feature selection, Random Forest / XGBoost classifier; SHAP explainability for top features
- Revenue Forecasting: Monthly revenue time series; SARIMA or Prophet model; 3-month forward forecast with confidence intervals
- Cohort Analysis: Monthly cohort retention matrix heatmap; customer lifecycle analysis; average order value by cohort
- Dashboard: Streamlit app with tabbed navigation — Overview, Segments, Churn, Forecast, Cohorts; interactive Plotly charts
- Report: Jupyter notebook-based HTML export + PDF executive summary with findings and business recommendations

■ 10-Day Milestone Plan

Day	Goals & Tasks
Day 1	Dataset acquisition, environment setup, JupyterLab config, MLflow tracking server, data dictionary documentation
Day 2	EDA Notebook — missing value audit, outlier detection, distribution plots, customer and product KPIs, correlation heatmap
Day 3	Data cleaning pipeline — missing value imputation, outlier capping, data type normalization; save cleaned dataset
Day 4	Feature engineering — RFM score computation, date-derived features, customer lifetime value approximation
Day 5	Customer segmentation — K-Means clustering, elbow and silhouette analysis, cluster labeling and business profiling
Day 6	Churn prediction — binary label engineering, Random Forest / XGBoost, cross-validation, ROC-AUC, SHAP feature importance

Day 7	Revenue forecasting — monthly time series construction, SARIMA/Prophet model, 3-month forecast, confidence intervals
Day 8	Cohort retention analysis — monthly cohort matrix, retention heatmap, average order value by cohort; statistical hypothesis testing
Day 9	Streamlit dashboard — tabbed navigation, Plotly charts for all analyses; model artifact logging to MLflow
Day 10	Executive BI report (findings + recommendations), Streamlit deployment (Streamlit Cloud), README, 5-min demo video

■ Deliverables

1. Jupyter notebooks — EDA, Feature Engineering, Segmentation, Churn, Forecasting, Cohort (all reproducible)
2. Cleaned and engineered dataset (CSV) with data dictionary
3. Trained model files — churn classifier (.pkl) and forecasting model (.pkl/.json) logged to MLflow
4. MLflow experiment tracking dashboard (local or hosted)
5. Interactive Streamlit dashboard (deployed on Streamlit Cloud)
6. Customer segmentation cluster profiles (visualization + business description per segment)
7. Churn model performance report (AUC, precision, recall, SHAP feature importance chart)
8. Revenue forecast visualization with 3-month prediction and confidence intervals
9. Cohort retention heatmap and lifecycle analysis
10. Executive BI Report (PDF) with findings, visualizations, and business recommendations

■ Evaluation Rubric

Criteria	Weight	Success Indicators
EDA & Data Quality	15%	Thorough EDA, data cleaning documented, data dictionary complete
Customer Segmentation	20%	RFM correct, K selection justified, cluster business profiles insightful
Churn Prediction Model	20%	AUC >80%, cross-validation correct, SHAP explainability clear
Revenue Forecasting	15%	SARIMA/Prophet correct, forecast confidence intervals included
Cohort Analysis	10%	Cohort matrix correct, retention trends interpreted correctly
Dashboard Quality	10%	Streamlit interactive, charts professional, navigation intuitive
Statistical Rigor	5%	Hypothesis tests applied correctly, p-values interpreted
Report & Presentation	5%	Business recommendations actionable, findings communicated clearly
TOTAL	100%	

■ Feature Checklist

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| ■ EDA: distribution plots for 10+ key metrics | ■ SARIMA or Prophet forecasting model |
| ■ Missing value and outlier audit with visual summary | ■ 3-month revenue forecast with confidence intervals |
| ■ RFM scoring (Recency, Frequency, Monetary) | ■ Monthly cohort retention matrix heatmap |
| ■ K-Means clustering with elbow and silhouette analysis | ■ Average order value by cohort analysis |
| ■ 4 customer segment business profiles | ■ Statistical hypothesis testing (t-test / chi-square) |
| ■ Churn binary label engineering from transaction history | ■ MLflow experiment and artifact tracking |
| ■ Random Forest / XGBoost churn classifier | ■ Streamlit dashboard with 5 tabbed analysis views |
| ■ SHAP feature importance waterfall and beeswarm plots | ■ Interactive Plotly charts throughout dashboard |
| ■ Monthly revenue time series construction | ■ PDF executive business intelligence report |

■ Learning Outcomes

- Conduct structured exploratory data analysis and document findings reproducibly
- Apply RFM analysis and K-Means clustering for enterprise customer segmentation
- Build and evaluate classification models with SHAP-based explainability for business audiences
- Implement time series forecasting using SARIMA or Prophet with confidence intervals
- Perform cohort analysis and retention modeling for customer lifecycle understanding
- Build interactive data dashboards using Streamlit and Plotly for executive audiences
- Track experiments and model artifacts using MLflow for reproducible data science
- Translate complex analytical findings into actionable business recommendations

■ Resources & References

- Pandas Docs: <https://pandas.pydata.org/docs/>
- scikit-learn: <https://scikit-learn.org/stable/documentation.html>
- Prophet Forecasting: <https://facebook.github.io/prophet/>
- statsmodels SARIMA: <https://www.statsmodels.org/stable/statespace.html>
- SHAP: <https://shap.readthedocs.io/>
- Streamlit Docs: <https://docs.streamlit.io/>
- MLflow: <https://mlflow.org/docs/latest/index.html>
- Plotly Python: <https://plotly.com/python/>
- Seaborn: <https://seaborn.pydata.org/>
- scipy.stats: <https://docs.scipy.org/doc/scipy/reference/stats.html>

■ Constraints & Guidelines:

- All notebooks must be fully reproducible — running cells top-to-bottom must produce identical results
- Random seeds must be fixed (random_state=42) in all stochastic operations

- K-Means cluster count must be justified with both elbow method and silhouette score — arbitrary selection is not acceptable
- Churn model must achieve AUC >0.80 on the held-out test set — do not report training set metrics as final performance
- SHAP explanations must be included for the churn model — black-box models without explainability are insufficient
- All visualizations must include axis labels, titles, and data source annotations
- Statistical hypothesis tests must state the null hypothesis, test statistic, and p-value interpretation
- The Streamlit dashboard must not reload the entire dataset on every interaction — use `st.cache_data` appropriately
- The executive BI report must include business recommendations — raw model metrics without interpretation are insufficient
- Model files must be serialized and logged to MLflow — do not submit only notebook outputs
- All code must include docstrings for non-trivial functions
- The dataset must have at least 10,000 transactions — toy datasets are not acceptable